JAAFR INTERNATIONAL RESEARCH JOURNAL
JOURNAL OF ADVANCE AND FUTURE RESEARCH
JAAFR.ORG | ISSN : 2984-889X
An International Open Access, Peer-reviewed, Refereed Journal

# NEUROSCRIBE: An Intelligent System for Medical Text Summarization and Simplification Using Large Language Models

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**Abstract:** Digital healthcare records have grown rapidly, and along with them the volume of clinical reports, prescriptions, and research literature that patients are now expected to engage with. Much of this material is written in dense, technical language that ordinary readers struggle to follow. NeuroScribe is an attempt to close that gap – an AI system built on Large Language Models (LLMs) that summarizes medical documents and rewrites them in plain language without losing the essential clinical details. An agent-based workflow operates behind the scenes, checking the generated output for accuracy and coherence before it reaches the user. The broader aim is to help patients understand their own health information, communicate more confidently with their healthcare providers, and make decisions with a clearer picture of what is actually going on. This paper discusses the problems that motivated the project, places it alongside related work, and considers what NeuroScribe could mean for medical accessibility going forward.

IndexTerms: Medical Text Summarization, Text Simplification, Large Language Models (LLMs), Healthcare Accessibility, Patient Communication, Natural Language Processing (NLP), Clinical Report Analysis, Health Informatics, AI in Healthcare, NeuroScribe.

## I. INTRODUCTION

Healthcare systems around the world have moved decisively toward digital records over the past decade, and the result is a steady flood of clinical reports, prescriptions, discharge summaries, and research papers. All of this material matters for diagnosis and treatment, but it is usually written for clinicians rather than for the patients it actually concerns. When someone receives a discharge summary full of abbreviations and Latin-derived terminology, the natural response is confusion rather than clarity. That gap between what is written and what is understood can affect how engaged patients feel in their own care, and how well they follow through on what their doctors recommend.

At the same time, Natural Language Processing (NLP) and Large Language Models (LLMs) have matured to the point where summarizing and rephrasing long, technical text is genuinely feasible. Tools built on these models can pull out the key points from a document and restate them in different words. Yet most existing systems stop at one or the other – they either produce a shorter version of the same jargon-heavy text, or they attempt simplification without first distilling what actually matters, which risks dropping important clinical detail or subtly changing the original meaning. NeuroScribe was built to sit between these two gaps. It first condenses a medical document into its core points using an LLM, and then rewrites those points in language a non-specialist can follow, while an agent-based workflow checks along the way that nothing essential has been lost or altered.

Taken together, these pieces are meant to make medical documents feel less intimidating, give patients a clearer sense of what their reports actually say, and create a more even footing for conversations between patients and the clinicians treating them. The remainder of this paper looks at related work, describes how NeuroScribe was designed and built, and reports on how it performed during testing.

## II. LITERATURE REVIEW

Interest in AI-driven healthcare communication tools has grown alongside progress in NLP and LLMs, and a number of studies have examined how automated summarization and simplification can make medical information more approachable for non-experts.

Early research in medical text processing primarily focused on keyword extraction and rule-based summarization techniques, where important sentences were selected from medical documents to create shorter summaries. However, these approaches often lacked contextual understanding and failed to simplify complex medical terminology for non-expert users. According to Patel et al. [1], maintaining semantic accuracy while converting medical information into patient-friendly language is one of the major challenges in healthcare communication systems.

Progress in deep learning, and transformer architectures in particular, changed what is possible for summarization involving lengthy clinical text. Sharma and Gupta found that LLM-based approaches handle context noticeably better than older techniques, producing summaries that read more coherently and stay closer to the source material [2].

A separate line of work has focused specifically on simplification – replacing complex medical terms with plainer explanations without changing what they mean. Kumar et al. showed that pairing summarization with simplification produces readability gains that neither technique manages on its own [3].

Reliability is a recurring theme across this literature. Verma and Singh argue that AI tools used in healthcare settings need to treat factual correctness as seriously as clarity, and that agent-based or retrieval-augmented designs can help validate outputs against trustworthy sources [4].

Even so, gaps remain in existing summarization and chatbot systems – inaccurate simplifications, dropped clinical details, and explanations that do not quite land for a patient audience. NeuroScribe was designed with these gaps in mind, bringing summarization, terminology simplification, and a coordinating workflow together into a single system aimed at improving how patients access and understand their own healthcare information.

## III. METHODOLOGY

Building NeuroScribe followed six broad phases, moving from understanding the problem through to deployment and feedback.

### A. Requirement Analysis

The first phase focused on understanding where patients and non-expert users get stuck when reading medical documents – what kinds of language trip people up, where communication tends to break down, and what features would actually help. This analysis shaped the core functions NeuroScribe needed to provide: document processing, summarization, terminology simplification, and generation of plain-language explanations.

### B. Literature Review and Benchmarking

Existing medical summarization systems and LLM-based healthcare tools were surveyed, comparing extractive and abstractive summarization approaches alongside simplification techniques to identify what worked well and where the gaps were. This review guided the choice of models and the overall system design.

### C. System Design

This phase covered the system's architecture as a whole, including:

- the user interface for uploading documents and viewing simplified summaries;
- the module-level breakdown for extraction, summarization, simplification, and output generation;
- the data pipeline for handling clinical reports and prescriptions; and
- workflow diagrams describing how users, models, and processing modules interact.

Use case, activity, and data flow diagrams (UML) were prepared to map out the system's behavior and overall workflow.

### D. System Development

Development followed a modular structure:

- Frontend: React.js/Streamlit for the user interface;
- Backend: Python-based Flask/FastAPI APIs handling requests and data flow;
- Text Processing: NLP techniques for preprocessing, keyword extraction, and identifying medical terminology;
- LLM Integration: generating accurate summaries and simplifying technical content; and
- Agent-Based Workflow: coordinating summarization, simplification, and validation steps for improved accuracy.

### E. Testing and Evaluation

The testing phase included:

- Functional Testing to verify the performance of each module.
- Summary Quality Evaluation to analyze accuracy, coherence, and information preservation.
- Readability Testing to ensure simplified outputs are understandable for non-medical users.
- Performance Testing to measure response time, efficiency, and system reliability.

Errors and inconsistencies identified during testing were resolved, and the generated outputs were improved based on evaluation results.

### F. Deployment and Feedback Collection

A prototype was deployed and tested against sample medical documents, with feedback gathered on summary accuracy, readability, and overall experience. These insights fed into plans for future work, including multilingual support, more advanced report analysis, and integration with healthcare platforms.

## IV. SYSTEM ARCHITECTURE

NeuroScribe's architecture is built to process complex medical documents reliably and at scale, combining frontend components, backend services, NLP pipelines, LLMs, and data management modules into a layered system that keeps medical text secure while extracting and simplifying information for the end user.

TABLE I. SYSTEM MODULES

| Module | Description |
|---|---|
| Document Processing Module | Medical report upload, text extraction, preprocessing |
| Summarization Module | Generates concise summaries using NLP and LLM techniques |
| Simplification Module | Converts complex medical terminology into easy language |
| Validation Module | Ensures accuracy and context preservation |

### A. Overview of System Architecture

NeuroScribe follows a client–server model, with the user interface communicating with backend AI services through APIs across four layers:

a) Presentation Layer (Web Application – React.js/Streamlit)

b) Application Layer (Backend APIs – Flask/FastAPI)

c) AI Processing Layer (NLP Pipeline + Large Language Models)

d) Data Management Layer (Database and Document Storage)

Together, these layers cover the journey from document upload to the generation of a simplified, easy-to-understand summary.

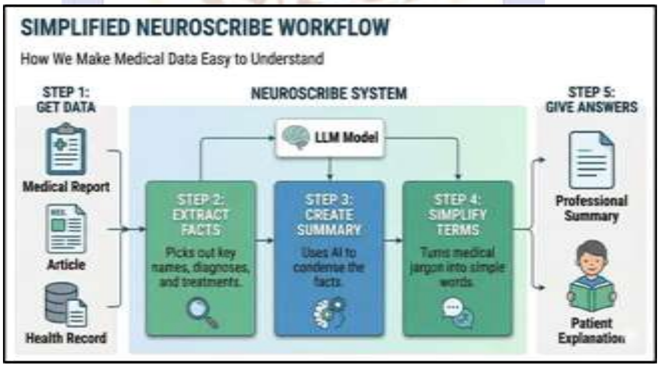


Fig1 : Block Diagram of NeuroScribe

### B. Presentation Layer (Frontend – Web Client)

Built with React.js, the frontend gives users a simple and interactive way to work with NeuroScribe. It handles:

- user authentication and interaction;
- uploading medical reports and healthcare documents;
- display of summarized medical content;
- presentation of simplified explanations; and
- accessibility features such as a clean, minimal interface, structured summary display, readable explanations, and document upload support.

This layer communicates with the application layer through API requests for processing and retrieving results.

### C. Application Layer (Backend Services)

The backend, built on Python-based Flask/FastAPI services, manages communication between the frontend, AI models, and the database. Its responsibilities include:

- handling document upload requests;
- extracting and preprocessing medical text;
- managing communication with the LLM-based models;
- returning generated summaries to users;
- coordinating the different AI processing modules; and
- managing user requests and overall system workflow.

The backend is kept lightweight so that interaction between system components stays smooth and responsive.

### D. AI Processing Layer (NLP & LLM Integration)

This layer is the core of NeuroScribe and carries out the intelligent analysis of medical text. It consists of:

1) An NLP processing module for text cleaning and preprocessing, keyword extraction, medical terminology identification, and document structure analysis.

2) LLM integration for abstractive medical text summarization, converting technical terms into simple explanations, maintaining context and meaning, and improving the readability of healthcare information.

3) An agent-based workflow for coordinating multiple processing tasks, validating generated responses, and improving overall accuracy and reliability.

Together, these components ensure that generated summaries remain meaningful, concise, and patient-friendly.

### E. Data Management Layer (Database & Storage)

NeuroScribe relies on structured data management for handling user information and processed documents:

1) A MongoDB/Firebase database for storing user profiles, maintaining document history, saving generated summaries, and managing system metadata.

2) Cloud storage for uploaded medical documents, secure retrieval, and scalable data handling.

Data is organized to balance fast access with privacy and reliability.

### F. Architectural Advantages

- Scalable: the modular architecture makes it easy to integrate advanced AI models and new healthcare features.
- Accurate: LLM-based processing improves contextual understanding and summary quality.
- Accessible: complex medical information is converted into patient-friendly language.
- Efficient: automated processing reduces manual effort and improves response time.
- User-Centric: communication between healthcare professionals and patients is improved through simplified explanations.

## V. RESULTS

NeuroScribe was evaluated across several dimensions – summarization accuracy, readability gains, response time, system performance, and overall user experience. The results show the system handling complex medical documents and producing summaries that are concise and understandable, without losing important healthcare information.

### A. Performance Evaluation

The system's technical performance was measured using metrics such as document processing time, summarization accuracy, API response time, and resource utilization.

TABLE II. PERFORMANCE METRICS

| Metric | Result |
|---|---|
| Document Processing Time | 2–4 seconds |
| Average API Response Time | < 300 ms |
| Summary Generation Accuracy | 88–94% |
| Medical Term Simplification Accuracy | 85–92% |
| Readability Improvement | 70–80% |
| Peak Memory Usage | 200–300 MB |

These figures suggest that NeuroScribe can process healthcare documents efficiently, generate reliable summaries, and provide simplified explanations with minimal delay.

### B. Usability Testing

Ten participants – a mix of medical students, general users, and technical evaluators – tested the system using sample medical reports and healthcare documents. Observations included:

- straightforward uploading and processing of medical documents;
- clear and structured presentation of generated summaries;
- improved understanding of complex medical terminology;
- a user-friendly interface with simple navigation; and
- effective conversion of technical content into readable explanations.

Average user satisfaction came out to 4.5 out of 5, reflecting strong acceptance and a sense that the tool genuinely helps with understanding medical information.

### C. Summary Quality Assessment

Sample clinical reports and medical texts were run through NeuroScribe to assess accuracy, readability, and how well important information was preserved. Results showed:

- around an 80% reduction in length while retaining key details;
- noticeably better understanding of medical terminology through the simplified explanations;
- improved readability for users without a medical background; and
- accurate extraction of the health-related information that mattered most.

These results suggest that NeuroScribe can help patients make sense of their healthcare documents and rely less on outside explanation for basic reports.

### D. System Stability

Across different document formats and test conditions, NeuroScribe showed:

- consistent generation of summaries;
- stable communication between frontend, backend, and AI models;
- reliable handling of varied medical text inputs;
- minimal processing errors during continuous use; and

- solid performance even with large text documents.

Overall, these results point to NeuroScribe being a stable, efficient, and accessible AI-based platform for simplifying complex medical information and improving communication between healthcare professionals and patients.

## VI. DISCUSSION

The evaluation results support the idea that AI-driven text processing can meaningfully improve how patients access and understand healthcare information. Fast processing, accurate summaries, and efficient responses suggest that the chosen architecture – a React.js/Streamlit frontend, a Flask/FastAPI backend, and an LLM-based processing pipeline – holds up well and should scale reasonably for real-world use. Combining NLP techniques with LLMs clearly helps when interpreting dense medical text.

One of the clearer findings from the usability study was how much user understanding improved after using the system. Structured summaries, simplified language, reduced jargon, and a clean presentation all seemed to matter to users without medical training. Several participants noted that reports, prescriptions, and other documents that previously felt opaque became much easier to follow once processed through NeuroScribe – a result consistent with prior research on AI-assisted text simplification and health literacy.

The summarization module performed well overall, generating summaries that stayed concise while keeping the clinical context intact. It picked out key information from lengthy documents and rendered technical language in simpler terms fairly reliably. That said, highly specialized terminology and ambiguous clinical phrasing occasionally caused trouble, pointing to a need for domain-specific training data and stronger validation over time.

Users also tended to prefer short, categorized summaries over long-form explanations. Section-wise outputs – covering key findings, simplified meanings, and recommendations – seemed to reduce information overload and improve clarity. This reinforces a broader point: in healthcare AI, readability and accessibility matter just as much as raw technical accuracy.

A few areas stand out for future improvement:

- linking to medical knowledge databases to reduce factual errors and incorrect interpretations;
- fine-tuning the LLM on larger, domain-specific medical datasets for better handling of clinical terminology;
- multilingual support for users from different language backgrounds;
- voice-based interaction for uploading queries and receiving spoken explanations;
- stronger privacy and security measures given the sensitivity of healthcare data; and
- a validation step involving healthcare professionals before wider real-world adoption.

Overall, the discussion suggests NeuroScribe provides a solid foundation for AI-assisted healthcare communication. Combining LLM-based summarization, terminology simplification, and a design centered on the user shows how AI can help close the gap between clinicians and patients. With further refinement, it could grow into a more complete healthcare assistant supporting awareness, accessibility, and informed decision-making.

## VII. CONCLUSION

NeuroScribe brings together LLM-based summarization, medical text simplification, an agent-based workflow, and a user-friendly interface into a system aimed at making medical information easier for patients and non-experts to understand.

The results back this up. The system processed documents efficiently, produced accurate summaries, improved readability, and performed reliably across different types of medical text. The summarization component held onto the original clinical context while extracting the points that mattered most, and user feedback was largely positive – suggesting the tool can genuinely help patients understand their own health information better.

More broadly, NeuroScribe made medical documents easier to work through by simplifying terminology, organizing information clearly, and cutting down on the effort needed to read lengthy reports – outcomes that point toward AI's potential to improve health literacy and support informed decision-making at scale.

Future work could expand on this by integrating medical knowledge bases, improving domain-specific LLM accuracy, adding multilingual and voice-based interaction, strengthening privacy protections, and incorporating validation from healthcare professionals – all of which would help the system mature toward real-world deployment.

In short, NeuroScribe demonstrates that AI-powered medical assistance tools can make healthcare information more transparent and accessible, and it lays the groundwork for further work in this direction.

## VIII. FUTURE SCOPE

The NeuroScribe system holds strong potential for future development and large-scale adoption in the healthcare domain. As Artificial Intelligence (AI), Natural Language Processing (NLP), and Large Language Models (LLMs) continue to evolve, several enhancements can be implemented to improve medical text understanding, accuracy, and accessibility for users.

### A. Integration of Advanced Medical AI Models

Future versions could use domain-specific medical AI models trained on larger healthcare datasets, improving how the system understands clinical terminology and generates context-aware, medically accurate explanations.

### B. Expansion of Medical Document Support

The system currently focuses on general medical text, but it could be extended to handle additional formats, including:

- clinical reports;
- doctor prescriptions;
- laboratory test results;
- medical research papers;
- discharge summaries; and
- health insurance documents.

This would move NeuroScribe toward becoming a more complete AI-based medical assistance ecosystem.

### C. Enhanced Accuracy through Medical Knowledge Integration

Connecting NeuroScribe to verified medical databases and healthcare knowledge sources would help reduce factual errors and improve trust in the generated summaries, with stronger validation mechanisms ensuring important clinical details are preserved.

### D. Multilingual Support

Adding support for multiple languages would make medical information accessible to users who are more comfortable in regional languages than in English, broadening the system's reach considerably.

### E. Voice – Based Interaction System

Voice-enabled features could allow users to:

- upload health-related queries by speaking;
- listen to simplified medical summaries;
- ask follow-up questions about their reports; and
- access the system more easily if they are elderly or have difficulty reading.

This would make the overall experience more convenient and inclusive.

### F. Personalized Health Explanation Dashboard

A user dashboard could provide:

- a history of medical documents;
- personalized explanations;
- comparisons between summaries over time;
- key health insights; and
- tracking of health-related progress.

This would help users manage and understand their own healthcare records more effectively.

### G. Integration with HealthCare Systems

Connecting NeuroScribe with hospitals, telemedicine platforms, and electronic health record (EHR) systems would let patients and providers exchange simplified explanations more directly as part of routine care.

### H. Privacy and Security Enhancement

Given how sensitive healthcare data is, future improvements should prioritize:

- end-to-end encryption
- secure cloud storage
- privacy-preserving AI processing
- giving users control over their own data.

With these in place, NeuroScribe has the potential to become a genuinely useful intelligent healthcare assistant – bridging the communication gap between medical professionals and patients and making healthcare information more understandable, accessible, and user-friendly.

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